Solution Requirements

Date	02/07/2026
Team ID	SWTID-2026-5714
Project Name	OptiCrop: Smart Agriculture Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

A comprehensive list of requirements defining exactly what the solution does and how well it performs. • Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

• Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Input Collection	The system shall process the submitted inputs through the trained Random Forest model and return the most suitable crop recommendation.	High
2	Crop Prediction	The system shall process the submitted inputs through the trained Logistic Regression model and return the most suitable crop recommendation.	High
3	Input Validation	The system shall validate that all required fields contain valid numeric values before submitting for prediction, and display an error message for invalid or missing input.	Medium
4	Result Display	The system shall display the predicted crop recommendation clearly on a result page immediately after prediction.	High

5	Navigation	The system shall provide Home, About, and Find Your Crop pages accessible through a common navigation menu.	Medium
6	Public Access	The system shall be accessible to any user via a public deployed URL without requiring installation, sign-up, or login.	High

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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system shall return the crop prediction result within 2 seconds of form submission on a standard internet connection.	Response time < 2 sec
2	Usability	The interface shall be simple enough for a first-time user to obtain a crop recommendation without needing instructions.	Task completed without external help
3	Reliability & Availability	The hosted application shall remain available for demonstration and grading, and shall handle invalid or missing input gracefully without crashing.	No crashes on invalid/missing input
4	Portability	The application shall run consistently across major web browsers and can be run locally on any machine with Python 3.10+ installed.	Works on Chrome, Firefox, Edge
5	Maintainability	The code shall be modular, separating the ML model, prediction logic, and UI templates for easy future updates.	Modular file structure (app.py, templates/, model files)
6	Accuracy	The trained model shall achieve a high test accuracy across all supported crop classes.	≥ 95% test accuracy (achieved: 97.3%)